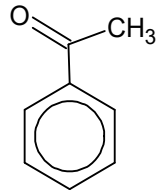
5 **(a)** Aluminium is a lightweight, strong and corrosion-resistant metal, widely used in aerospace, construction, packaging and electrical transmission. Its compounds are important in manufacturing, water treatment and pharmaceuticals. One such compound is anhydrous aluminium chloride which catalyses the synthesis of acetophenone from benzene, which is similar to Friedel-Crafts alkylation.



Acetophenone

(i) Write an equation to show how aluminium chloride functions as a Lewis acid catalyst in the synthesis of acetophenone by using a suitable acyl chloride. **[1]**

(ii) Explain, with the aid of an equation, why aluminium chloride must be anhydrous in order for it to function as a Lewis acid catalyst during the synthesis of acetophenone from benzene. **[2]**

(iii) Describe the mechanism for the synthesis of acetophenone from benzene, showing curly arrows, charges and any relevant lone pairs. **[2]**

(iv) Suggest a simple chemical test to confirm the presence of acetophenone. **[2]**

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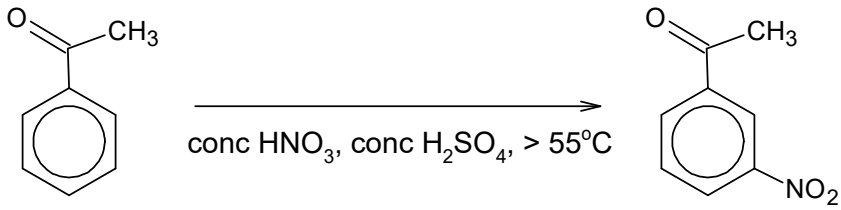
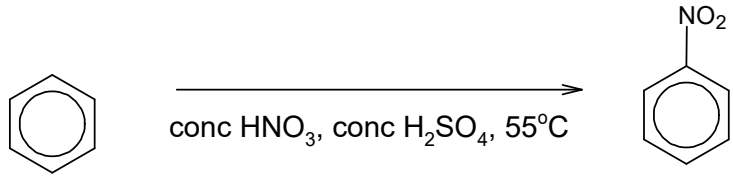
(b) **(i)** Both acetophenone and benzene react with nitric acid, but under different conditions to form compounds **S** and **T** respectively.



S



T

Explain why different conditions are needed for these two reactions. **[2]**

(ii) The solubility of **S** and **T** in water are shown in Table 5.1.

Table 5.1

Compound	Solubility in water / mg dm^{-3}
S	108
T	80

Explain, in terms of structure and bonding, the difference in solubility between **S** and **T**. **[2]**

(c) **E**, **F** and **G** are three elements in Period 3 of the Periodic Table. The pH of the solutions of their oxides and chlorides are as shown in Table 5.2.

Table 5.2

	E	**F**	**G**
pH of oxides	9	7	2
pH of chlorides	6	3	2

(i) Identify the elements **E**, **F** and **G**. **[1]**

(ii) Explain your choice of the elements in **(c)(i)** **[3]**

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(d) In the gaseous phase, aluminium chloride dimerises to form Al_2Cl_6, as shown in the equation below.

$$2\ AlCl_3\ (g) \rightarrow Al_2Cl_6\ (g) \qquad \Delta H_{dimerisation} = -158.8\ kJ\ mol^{-1}$$

(i) Draw the structure of the dimer, Al_2Cl_6 and state the shapes around *Al* and *Cl* atom. **[2]**

(ii) The average bond energy for the *Al*–*Cl* bond in Al_2Cl_6 can be determined using the relevant data in the *Data Booklet* and the following data.

Enthalpy change of dimerisation of Al_2Cl_6 (g) = –158.8 kJ mol^{-1}
Enthalpy change of formation of $AlCl_3$ (g) = – 248.6 kJ mol^{-1}
Enthalpy change of atomisation of *Al* (s) = + 326 kJ mol^{-1}

Construct an energy cycle to calculate the average bond energy for the *Al*–*Cl* bond in Al_2Cl_6. **[3]**

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